

Problem Statement: Web Scraping Flipkart Smartphone Listings

Objective:

Write a Python program to parse a saved HTML page (flipkart.html) containing Flipkart smartphone listings and extract structured data for analysis.

Input

- A saved Flipkart HTML file: flipkart.html (Load file using 'utf-8' encoding)
 - The file contains multiple smartphone listings with product details such as name, specifications, price, and reviews.
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Tasks

You must complete the following:

1. HTML Parsing

- Load and parse the flipkart.html file using BeautifulSoup.

2. Product Block Extraction

- Identify each product's container or listing block.

3. Field Extraction

For each product, extract the following fields:

Field	Description
Name	Full name of the phone including brand
Memory	Storage capacity (e.g., 128 GB)
Color	Color variant of the phone (e.g., Dark Chrome)
Stars	Star rating (e.g., 4.5)
Reviews	Number of user reviews
Actual Price	Original price before discount (if available) (₹ symbol to be removed)
Final Price	Price after applying discount (₹ symbol to be removed)
Exchange Offer	Value of exchange offer (if available) (₹ symbol to be removed)
Discount	Percentage discount (if displayed, like "10%")

 If any field is missing for a product (e.g., no exchange offer, no discount, no actual price, etc), it should be filled with NaN.

4. Data Storage

- Store the extracted data in **Pandas DataFrame**.

5. Output

- Display the created DataFrame with all the rows.
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Constraints

- Use only local file parsing with **BeautifulSoup**, any other method is not allowed.
 - Focus on **robustness**: handle missing fields cleanly.
 - Name of the columns should be same as shown in the sample output.
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Expected Output

- A preview of all products with all extracted fields shown clearly (as a dataframe), The sample output with first 16 rows is shown below:

	Name	Memory	Color	Stars	Reviews	Actual Price	Final Price	Exchange Offer	Discount
0	Nothing Phone (3a) Pro	128 GB	Black	4.5	439	32,999	29,999	21,800	9%
1	Nothing Phone (3a)	128 GB	Black	4.5	2,053	27,999	24,999	18,250	10%
2	OnePlus Nord CE4	256 GB	Celadon Marble	4.4	1,267	26,999	23,999	NaN	11%
3	OnePlus Nord CE4 lite 5G	128 GB	MEGA BLUE	4.4	1,032	20,999	20,990	NaN	NaN
4	OnePlus 13R 5G	512 GB	Nebula Noir	4.5	39	51,999	43,174	NaN	16%
5	OnePlus Nord CE4	128 GB	Dark Chrome	4.4	1,267	24,999	21,987	NaN	12%
6	OnePlus Nord CE4	128 GB	Celadon Marble	4.4	1,267	24,999	21,899	NaN	12%
7	OnePlus Nord CE4 lite 5G	256 GB	MEGA BLUE	4.4	1,032	23,999	20,800	NaN	13%
8	OnePlus Nord CE4 lite 5G	256 GB	SUPER SILVER	4.4	1,032	23,999	21,350	NaN	11%
9	OnePlus 13R 5G	256 GB	Nebula Noir	4.5	134	44,999	42,990	NaN	4%
10	Samsung Galaxy A35 5G	128 GB	Awesome Iceblue	4.3	922	33,999	22,999	NaN	32%
11	Samsung Galaxy M35 5G	128 GB	DayBreak Blue	4.4	1,674	NaN	24,499	NaN	NaN
12	Samsung Galaxy M35 5G	128 GB	Thunder Grey	4.4	1,674	24,499	24,498	NaN	NaN
13	Samsung Galaxy A35 5G	128 GB	Awesome Lilac	4.3	922	33,999	22,999	17,150	32%
14	Apple iPhone 13	128 GB	Starlight	4.6	13,917	49,900	44,999	33,250	9%
15	Samsung Galaxy M35 5G	256 GB	DayBreak Blue	4.3	151	27,779	22,999	NaN	17%

